

JIANUO QIU

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EDUCATION

Georgia Institute of Technology

M.Sc in Robotics

Overall GPA: 4.00/4.00

Aug 2025 - Present

Atlanta, GA

Southern University of Science and Technology

B.Eng in Robotics Engineering

Weighted Average: 89.5/100.0 | Overall GPA: 3.69/4.00 | Major GPA: 3.84/4.00

Aug 2021 - Jun 2025

Shenzhen, China

PUBLICATION

- Han, Y.* , **J. Qiu***, L. Bai, Z. Xiao, Z. Zeng, Y. Liu, Z. Yang, S. Jain, W. Ma, J. Fu, Y. Zheng, M. Natarajan, M. Z. Irshad, K. Shaw, M. Gombolay, Z. Kira, and H. Ravichandar, “Video2Sim2Real: Full-Stack Autonomous Dexterous Skill Acquisition from a Single Human Video,” *arXiv preprint arXiv:2606.08828*, 2026.
- Zhang, R.* , W. Qiu*, **J. Qiu**, Y. Guo, C. Dong, T. Zhang, J. Yi, C. Song, H. Asada, and F. Wan, “Multi-Modal Intention Recognition Combining Head Motion and Throat Vibration for Underwater Superlimbs,” *IEEE Transactions on Automation Science and Engineering*, 2025.
- Chen, Y.* , Zhang, C.* , P. Gu* , **J. Qiu**, J. Yin, N. Qiu, G. Huang, B. Huang, Z. Zhang, H. Deng, W. Zhang, F. Wan, and C. Song, “Evolutionary Morphology Towards Overconstrained Locomotion via Large-Scale, Multi-Terrain Deep Reinforcement Learning,” in *Proceedings of the 6th IEEE International Conference on Reconfigurable Mechanisms and Robots (ReMAR)*, 2024, pp. 401–408.

RESEARCH EXPERIENCE

Video2Sim2Real

Core Member, the First Author

Dec 2025 - Jun 2026

Atlanta, GA

- Developed an object-centric keyframe refinement module that optimizes robot configurations using object geometry and motion, and interpolates refined trajectories for dexterous manipulation from a single human video.
- Designed and trained distillation imitation-learning policies that maps noisy object point clouds to calibrated keyframe hand poses, improving robustness to local object-pose variations during sim-to-real deployment.
- Implemented DP3-based imitation-learning baselines and evaluated them against the proposed decoupled IL + residual RL strategy in both simulation and real-world settings.
- Led all real-world experiments on a Kinova Gen3 arm with a LEAP Hand, including evaluations of the proposed method, sim-to-real baselines, and policy ablations across everyday dexterous manipulation tasks.

Spherical Linkage as Limbs in Flapping-Wing Robots

Undergraduate Project

Jun 2024 - Jun 2025

Shenzhen, China

- Designed robot wings in Fusion 360 with three kinds of configurations: planar, spherical, and Bennett 4-bar linkages, and fabricated wing prototypes with several sets of design parameters for each configuration

- Conducted aerodynamics simulation experiments with static wings in ANSYS Fluent to estimate lift force and drag force, and evaluated performance across different wing configurations under different wind velocities
- Built an experimental platform with a 6-axis force/torque sensor, conducted flapping-wing experiments, and collected lift force data
- Developed and trained neural network models for different wing configurations taking design parameters and flapping frequency as input and lift force maximum as output to estimate optimal design parameters, and validated the effectiveness of a hierarchical training strategy combining pretraining and fine-tuning for neural networks.

Overconstrained 4-bar Linkages as Robot Limbs in Locomotion

Nov 2023 - Apr 2024

Core Member, the Second Author

Shenzhen, China

- Trained wheel-legged overconstrained locomotion models using deep reinforcement learning in Isaac Gym across diverse terrains, including discrete flat surfaces, rough slopes, uphill, and downhill environments; fine-tuned reward parameters to optimize locomotion performance
- Extracted simulation data including torque, velocity, and distance from Isaac Gym and computed the Cost of Transport (CoT) for forward and lateral locomotion. The overconstrained limb demonstrated 22.4% and 30.6% lower energy consumption than the planar limb during forward and sideways walking, respectively, confirming its superior energy efficiency.

Multi-modal Intention Recognition for Underwater Superlimb

Aug 2023 - Sep 2024

Core Member, the Second Author

Shenzhen, China

- Conducted underwater intention-recognition and wearable interaction experiments, and collected multimodal datasets for training and testing
- Developed a MATLAB-based GUI for both forward and inverse force analysis of two thrusters, supporting 3 kinds of input methods including joystick control, mouse interaction, and superlimb input
- Extracted force data from the superlimb, and visualized real-time data within the MATLAB GUI

'Kunyi'-Underwater Reconfigurable Robot

Oct 2022 - Nov 2022

Core Member

Shenzhen, China

- Conducted force analysis and verification of shaft components
- Presented the work and received First Prize in the 15th International Robot & Simulation Technology Competition

LEADERSHIP EXPERIENCE

Mutual Aid Class, Zhixin College, SUSTech

Apr 2023 - Jun 2024

Peer Tutor

Shenzhen, China

- Selected by Zhixin College as a peer tutor to assist undergraduates in reviewing the course *Probability and Statistics*
- Mentored a class of 15 students, clarifying key concepts and problem-solving techniques

HONORS & AWARDS

National Encouragement Scholarship, China

2024

Outstanding Student Scholarship, SUSTech

2024

First Prize, 15th International Robot & Simulation Technology Competition

2022

TECHNICAL SKILLS

Programming Languages
Software

Python, MATLAB, C++, C#, Java, LaTeX

Isaac Gym, SolidWorks, AutoCAD, Fusion 360, ANSYS Fluent